FR3 WIDEBAND ARRAY WITH BEAM-SQUINT MITIGATION

FOR VEHICULAR 6G INTEGRATED SENSING AND COMMUNICATION (ISAC)

PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that the project report entitled “FR3 Wideband Array with Beam-Squint Mitigation for Vehicular 6G Integrated Sensing and Communication (ISAC)” is a bonafide record of project work carried out by the students named below under the guidance and supervision of the faculty for BECE305L – Antennas and Microwave Engineering.

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The report presents an analytical MATLAB investigation of wideband array beam steering and beam-squint mitigation. The preliminary HFSS single-element and eight-element array work is described with its current validation status and is not represented as a converged electromagnetic result.

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ABSTRACT

Wideband phased arrays are important for emerging sixth-generation (6G) vehicular systems, in which Integrated Sensing and Communication (ISAC) uses shared radio hardware and spectrum to support both data links and environmental sensing. A key limitation of conventional phase-shift beamforming is beam squint: a phase progression selected at one frequency does not preserve the same steering direction across a wide frequency band. This project studies that effect in an FR3-oriented array and evaluates true-time-delay (TTD) as a mitigation technique.

A rectangular microstrip patch element was analytically synthesized for 7.5 GHz using the transmission-line/cavity-model approach for a Rogers RT/duroid 5880 substrate (relative permittivity 2.2, thickness 1.575 mm). The element geometry was checked by recomputing its resonant frequency. An eight-element uniform linear array was modelled in MATLAB with half-wavelength spacing at the highest test frequency. Conventional fixed-phase steering, ideal TTD steering and a 5-bit quantized TTD implementation were compared over 7.0–8.5 GHz at a 25° target direction.

The MATLAB study reports a 4.50° beam-angle spread for fixed-phase steering, compared with 0.25° for ideal TTD and 0.55° for quantized 5-bit TTD. At a 50 m reference range, the reported worst-case position-error magnitude was 3.41 m for fixed-phase steering, 1.05 m for ideal TTD and 1.22 m for quantized TTD. The corresponding reported worst-case communication gain losses were approximately 0.922 dB, 0.058 dB and 0.079 dB. These results indicate that TTD substantially reduces frequency-dependent pointing variation in the analytical model. In HFSS, an inset-fed single patch showed an S11 minimum of −18.31 dB at 7.4325 GHz with a simulated peak gain of 7.59 dB, and an eight-element patch array showed active-S minima of about −10 to −17 dB around 7.8–7.95 GHz. An HFSS cross-check of the eight-element array, using the same fixed and frequency-scaled port phases, gave a peak-direction drift of 5.6° for fixed phase and 1.6° for TTD phases over 7.0–8.5 GHz, following the MATLAB trend. The HFSS adaptive passes did not meet the requested convergence criterion, so the full-wave results are reported as preliminary.

Keywords: FR3, 6G, vehicular ISAC, microstrip patch, phased array, beam squint, true-time delay, MATLAB.

1. INTRODUCTION

The evolution toward 6G wireless systems is expected to increase demand for high-throughput, low-latency and context-aware connectivity. Vehicular communication is particularly demanding because vehicles exchange information with nearby vehicles and roadside infrastructure while also sensing objects, road conditions and surrounding traffic. Integrated Sensing and Communication (ISAC) seeks to use common radio resources for communication and sensing, potentially reducing duplicated hardware and enabling closer coordination between the two functions.

The FR3 region, commonly used to describe upper-mid-band spectrum around 7–24 GHz in current 6G discussions, offers a potential balance between propagation characteristics and wider bandwidth availability. Directional antenna arrays can provide beamforming gain and electronic steering, but wideband operation exposes a limitation of conventional phase-shifter-based steering: beam squint.

1.1 Problem statement and motivation

In a conventional phased array, the phase offset between adjacent elements is calculated at a design frequency. When operating frequency changes, the same phase offset corresponds to a different time delay relative to wavelength. The main radiation lobe therefore shifts away from the commanded direction. In vehicular ISAC, this frequency-dependent pointing error can reduce received communication power and introduce angular error into sensing-based position estimates.

This project investigates the effect using an eight-element linear array and compares fixed-phase steering with ideal and quantized true-time-delay steering across a 7.0–8.5 GHz sweep. The central question is not only whether the beam moves, but how much that movement affects representative sensing and communication metrics.

1.2 Objectives and goals

- Analytically synthesize a rectangular microstrip patch element at 7.5 GHz on Rogers RT/duroid 5880.
- Compute the element radiation pattern using a two-slot cavity-model approximation.
- Build an eight-element uniform linear array and evaluate its frequency-dependent radiation pattern in MATLAB.
- Compare fixed-phase steering, ideal TTD steering and 5-bit quantized TTD steering for a 25° target angle.
- Translate pointing error into a 50 m reference-range position-error metric and calculate link-gain loss at the target bearing.
- Document modelling assumptions and distinguish completed analytical results from the still-unvalidated HFSS single-element simulation.
- Cross-check the beam-squint trend in an eight-element HFSS model driven with fixed and frequency-scaled port phases.

1.3 Applications and features

Potential applications include vehicle-to-vehicle links, vehicle-to-infrastructure communication, radar-assisted object localization, road-hazard detection and joint communication/sensing beam management. Key features of the study are a reproducible frequency sweep, a common array geometry for all steering methods, explicit modelling of finite phase resolution, and paired sensing/communication metrics.

The work is an analytical and MATLAB-based study rather than a claim of a fully fabricated or electromagnetically validated array. Its value lies in isolating beam-squint behaviour and quantifying the benefit of TTD under controlled assumptions.

2. THEORY AND DESIGN PARAMETERS

2.1 Beam squint in wideband arrays

For an N-element uniform linear array, the array factor is the coherent sum of contributions from individual elements. A general form is:

$$AF(\theta,f) = \Sigma[n=0\ldots N-1]\ w_n(f)\ \exp\{j[n\ k(f)d\ \sin\theta + \psi_n(f)]\}$$

Here, N is the number of elements, d is the inter-element spacing, $k(f)=2\pi f/c$ is the free-space wavenumber, w_n is the amplitude weight and ψ_n is the excitation phase. The exact sign of the phase term depends on the adopted convention; steering and observation definitions must be consistent.

For fixed-phase steering, the phase progression is calculated at reference frequency f_0 and held constant. Because k(f) changes with frequency, the peak direction no longer remains fixed. This is beam squint [1], [2], [4].

2.2 True-time-delay steering

A true-time-delay network applies a physical delay τ_n to each element. The corresponding phase is frequency dependent, $\varphi_n(f)=-2\pi f\tau_n$ (the sign depends on convention). Since phase scales with frequency, the intended time-delay progression can preserve the target direction over a wider bandwidth, which is critical for joint communications and radar sensing in 6G systems [3], [5]. Practical delay networks and quantized phase controls still introduce residual error.

The ideal TTD case uses continuous delay values. The 5-bit case models a 32-state phase-control resolution, equivalent to 11.25° per step, by rounding the ideal phase to the nearest allowed state.

2.3 Patch synthesis and array parameters

PARAMETER	VALUE	RATIONALE
Substrate	Rogers RT/duroid 5880, ε_r=2.2, h=1.575 mm	Low-loss microwave substrate
Design frequency	7.5 GHz	Analytical patch design point
Sweep	7.0–8.5 GHz; 0.25 GHz step	Wideband beam-squint evaluation
Array size	8-element uniform linear array	Compact demonstrator
Element spacing	λ/2 at 8.5 GHz ≈17.635 mm	Avoid grating lobes across sweep
Target steering angle	25° from broadside	Off-broadside vehicular link
Phase-shifter resolution	5-bit; 32 states; 11.25°	Finite-resolution comparison
Reference range	50 m	Illustrative sensing metric

The analytical patch synthesis produced W = 15.800 mm, physical length L = 12.488 mm, fringing extension ΔL = 0.814 mm per edge, effective length L_e = 14.115 mm and effective dielectric constant εeff = 2.0049. Recalculation from the effective length reproduced 7.5000 GHz as the self-consistency check.

The array spacing was selected at the highest swept frequency so that the spacing did not exceed half a wavelength over the sweep, reducing the risk of grating lobes in the model.

3. DESIGN AND IMPLEMENTATION

3.1 Overall workflow

1. Define the design point, substrate and operating-frequency sweep.
2. Calculate patch width, effective permittivity, fringing extension and effective length using standard patch synthesis equations.
3. Calculate the patch element factor from the two-slot cavity model.
4. Create an eight-element linear-array model and set the spacing to approximately 17.635 mm.
5. Calculate steering phases for the 25° target using fixed-phase, ideal TTD and 5-bit quantized TTD strategies.
6. At each frequency, evaluate the total pattern on an angular grid from −90° to +90° and find its peak direction.
7. Calculate pointing error, reference-range position error and gain loss at the intended target bearing.
8. Compare the three steering strategies over the same seven frequencies.

3.2 Modelling assumptions

- The total pattern is obtained by multiplying the array factor by the single-element H-plane pattern.
- Mutual coupling between array elements is neglected.
- The patch is represented by an analytical cavity-model element pattern; measured or full-wave element data are not yet substituted.
- The sensing illustration uses a fixed reference range of 50 m. It is a geometry-based error indicator, not a complete radar tracker simulation.
- Communication gain loss is evaluated relative to the peak of the same array configuration and expressed in decibels at the target bearing.

3.3 ISAC performance metrics

For a small angular pointing error $\Delta\theta$ at reference range R, the transverse position-error magnitude is approximated by $|e_p| = R \tan(|\Delta\theta|)$. The sign can be retained separately to indicate the direction of angular deviation; the magnitude is the relevant measure for localization impact.

Link-gain loss is the difference between peak array gain and the gain delivered at the true 25° bearing. A small value indicates that the target remains close to the main-lobe peak even if the exact peak direction is slightly offset.

3.4 MATLAB implementation structure

The MATLAB workflow is organized into parameter initialization, patch synthesis, element-pattern generation, array-factor evaluation, three steering strategies, peak-angle extraction and metric calculation. Frequency points are processed with a common loop so the comparison uses identical frequency and angular grids.

3.5 MATLAB IMPLEMENTATION — CORE SCRIPT

Listing 3.1: Illustrative MATLAB script excerpt for array factor and beam-squint calculation (Note: This illustrative excerpt outlines the array-factor kernel; in the full numerical analysis, total pattern multiplication with the normalized element factor is incorporated to yield the 23.9° peak at 7.5 GHz).

The following code is a compact implementation outline of the analytical workflow. Keep the phase sign and angle convention consistent with the project plots when running or editing it. The numerical results reported in Section 4 are the values recorded in the supplied project analysis.

```
clear; clc; close all;

% FR3 array: beam-squint and TTD comparison
c = 3e8;
er = 2.2; h = 1.575e-3; fc = 7.5e9;
freq = (7.0:0.25:8.5)*1e9;
N = 8; theta0 = 25; R = 50;
bits = 5; phaseStep = 360/(2^bits);

% Rectangular patch synthesis (transmission-line model)
W = c/(2*fc)*sqrt(2/(er+1));
eeff = (er+1)/2 + (er-1)/2*(1+12*h/W)^(-0.5);
dL = 0.412*h*((eeff+0.3)*(W/h+0.264)) / ...
     ((eeff-0.258)*(W/h+0.8));
Le = c/(2*fc*sqrt(eeff));
L = Le - 2*dL;
fr_check = c/(2*(L+2*dL)*sqrt(eeff));
```

```
fprintf('W = %.3f mm, L = %.3f mm\n',W*1e3,L*1e3);
fprintf('eeff = %.4f, dL = %.3f mm\n',eeff,dL*1e3);
fprintf('Le = %.3f mm, fcheck = %.4f GHz\n', ...
        Le*1e3,fr_check/1e9);

% Array geometry and angular grid
d = c/(2*max(freq));
n = (0:N-1).';
theta = -90:0.05:90;
k0 = 2*pi*fc/c;
psi0 = -k0*d*sind(theta0)*n;
tauStep = d*sind(theta0)/c;

% Simplified normalized element factor for array modelling
elementFactor = abs(cos(pi*(W/(c/max(freq))).*sind(theta)));
elementFactor = elementFactor ./ max(elementFactor);

beamFixed = zeros(size(freq)); beamTTD = beamFixed;
beamQTTD = beamFixed; lossFixed = beamFixed;
lossTTD = beamFixed; lossQTTD = beamFixed;
posFixed = beamFixed; posTTD = beamFixed; posQTTD = beamFixed;
```

3.6 MATLAB IMPLEMENTATION — FREQUENCY SWEEP

```
for m = 1:numel(freq)
    f = freq(m); k = 2*pi*f/c;
    afFixed = zeros(size(theta));
    afTTD = zeros(size(theta)); afQTTD = zeros(size(theta));

    % Ideal delay and 5-bit phase quantization
    psiTTD = -2*pi*f*tauStep*n;
    psiQTTD = round(psiTTD/(phaseStep*pi/180)) * ...
              (phaseStep*pi/180);

    for q = 1:numel(theta)
        spatial = k*d*sind(theta(q))*n;
        afFixed(q) = abs(sum(exp(1j*(spatial + psi0))));
        afTTD(q) = abs(sum(exp(1j*(spatial + psiTTD))));
        afQTTD(q) = abs(sum(exp(1j*(spatial + psiQTTD))));
    end

    AF1 = afFixed/max(afFixed);
    AF2 = afTTD/max(afTTD);
    AF3 = afQTTD/max(afQTTD);
    [~,i1] = max(AF1); [~,i2] = max(AF2); [~,i3] = max(AF3);
    beamFixed(m) = theta(i1);
    beamTTD(m) = theta(i2);
    beamQTTD(m) = theta(i3);

    [~,it] = min(abs(theta-theta0));
    lossFixed(m) = -20*log10(max(AF1(it),eps));
    lossTTD(m) = -20*log10(max(AF2(it),eps));
    lossQTTD(m) = -20*log10(max(AF3(it),eps));

    posFixed(m) = R*tand(beamFixed(m)-theta0);
    posTTD(m) = R*tand(beamTTD(m)-theta0);
    posQTTD(m) = R*tand(beamQTTD(m)-theta0);
end

figure; plot(freq/1e9,beamFixed,'o-',freq/1e9,beamTTD,'s-', ...
    freq/1e9,beamQTTD,'^-','LineWidth',1.3);
grid on; xlabel('Frequency (GHz)'); ylabel('Beam angle (deg)');
legend('Fixed phase','Ideal TTD','5-bit TTD','Location','best');
title('Main-beam direction versus frequency');

figure; plot(freq/1e9,abs(posFixed),'o-',freq/1e9,abs(posTTD),'s-', ...
    freq/1e9,abs(posQTTD),'^-','LineWidth',1.3);
grid on; xlabel('Frequency (GHz)'); ylabel('Position-error magnitude (m)');
legend('Fixed phase','Ideal TTD','5-bit TTD','Location','best');

figure; plot(freq/1e9,lossFixed,'o-',freq/1e9,lossTTD,'s-', ...
    freq/1e9,lossQTTD,'^-','LineWidth',1.3);
grid on; xlabel('Frequency (GHz)'); ylabel('Gain loss (dB)');
legend('Fixed phase','Ideal TTD','5-bit TTD','Location','best');
```

3.7 HFSS simulation status

An inset-fed rectangular patch was also explored in Ansys HFSS (Student 2025 R2.4), using Rogers RT/duroid 5880 and a 7.5 GHz target, with a lumped port at the feed line and perfect-E boundaries on the patch and ground plane (Figure 3.1). The analytical transmission-line synthesis yields an ideal patch length of 12.488 mm. In the HFSS model, parametric simulation tuning was performed to optimize input impedance match with the 50 Ω inset feed line on the 1.575 mm substrate, resulting in a tuned patch length of 16.0 mm, patch width 15.81 mm, feed-line width 4.8 mm, inset depth 4 mm and inset gap 0.8 mm, with approximately 11 mm of air-box padding around the substrate.

The simulated S11 of the single patch, swept from 6.0 to 9.0 GHz, reaches a minimum of −18.31 dB at 7.4325 GHz (Figure 3.2), about 0.9% below the 7.5 GHz design frequency. Reading the curve, S11 stays below −10 dB over roughly 7.3–7.6 GHz, i.e. an impedance bandwidth of about 0.25 GHz (graphical estimate). The Message Manager still reports that the adaptive passes did not converge to the requested ΔS criterion, although the interpolating frequency sweep itself converged (31 points). The HFSS results are therefore presented as simulation evidence with this caveat, and are kept separate from the analytical MATLAB array study.

The single patch was extended to eight elements along the X axis with 17.647 mm spacing (λ/2 at 8.5 GHz; MATLAB uses 17.635 mm). The substrate (148.79 × 24.95 × 1.575 mm) and air box (170.79 × 46.95 × 23.575 mm, about 11 mm clearance) were enlarged to enclose the array. Each element has its own lumped port (modal solution, eight ports). The sweep covers 7.0–8.5 GHz with 31 interpolating points (solution frequency 7.5 GHz, maximum 7 passes, ΔS = 0.02). Far-field patterns at 7.0, 7.5, 8.0 and 8.5 GHz came from a discrete sweep with fields saved.

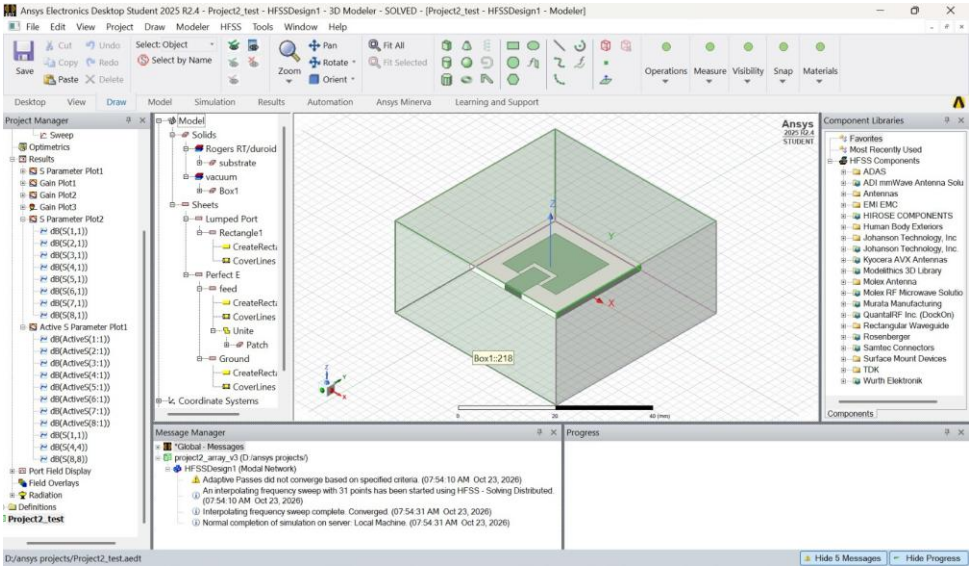


Figure 3.1: HFSS 3D model of the inset-fed single patch on Rogers RT/duroid 5880 with lumped port and air box.

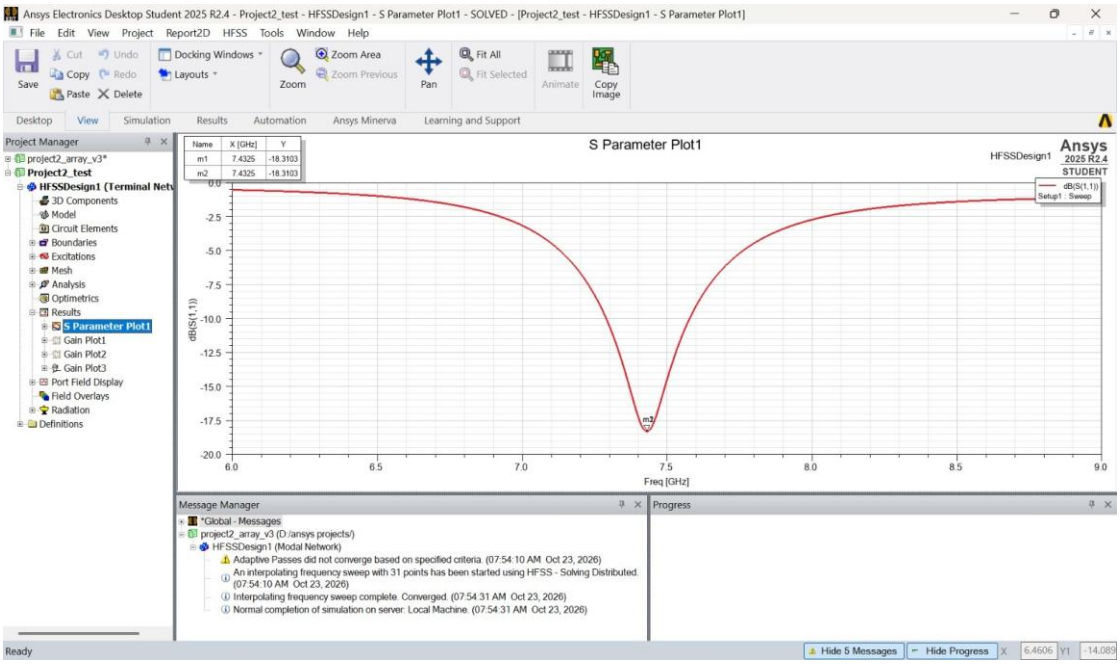


Figure 3.2: Simulated S11 of the single patch (6–9 GHz); minimum −18.31 dB at 7.4325 GHz (markers m1/m2).

4. RESULTS AND DISCUSSION

4.1 Synthesized patch geometry

The analytical patch synthesis produced the following dimensions. The self-consistency check returned the intended design frequency of 7.5000 GHz.

Table 4.1: Synthesized rectangular microstrip patch antenna dimensions at 7.5 GHz

QUANTITY	SYMBOL	VALUE
Width	W	15.800 mm
Physical length	L	12.488 mm
Fringing extension per edge	ΔL	0.814 mm
Effective length	L_e	14.115 mm
Effective dielectric constant	εeff	2.0049
Self-check resonance	f_r,check	7.5000 GHz

4.2 Beam-angle comparison

Across seven frequency points from 7.00 to 8.50 GHz, fixed-phase steering produced a beam-angle range of 25.60° to 21.10°, a 4.50° spread. Ideal TTD remained within 23.80°–24.05°, a 0.25° spread. Quantized 5-bit TTD remained within 23.60°–24.15°, a 0.55° spread. The quantized case retained most of the benefit of ideal TTD while representing finite phase resolution.

Table 4.2a: Summary of beam steering angle range and spread (MATLAB analytical array)

STEERING METHOD	BEAM ANGLE RANGE	SPREAD	INTERPRETATION
Fixed phase	21.10°–25.60°	4.50°	Largest frequency-dependent drift
Ideal TTD	23.80°–24.05°	0.25°	Most stable steering
5-bit TTD	23.60°–24.15°	0.55°	Small quantization effect

Table 4.2: Full steering and ISAC metric results across the frequency sweep

Freq (GHz)	Phase-shift angle (deg)	Phase-shift error (deg)	Phase-shift pos. error (m)	Phase-shift gain loss (dB)	TTD angle (deg)	TTD error (deg)	TTD pos. error (m)	TTD gain loss (dB)	TTD 5-bit angle (deg)	TTD 5-bit error (deg)	TTD 5-bit pos. error (m)	TTD 5-bit gain loss (dB)
7.00	25.60	0.60	0.52	0.015	23.80	-1.20	-1.05	0.058	23.60	-1.40	-1.22	0.079
7.25	24.75	-0.25	-0.22	0.003	23.85	-1.15	-1.00	0.057	23.75	-1.25	-1.09	0.070
7.50	23.90	-1.10	-0.96	0.055	23.90	-1.10	-0.96	0.055	24.05	-0.95	-0.83	0.042
7.75	23.15	-1.85	-1.62	0.172	23.95	-1.05	-0.92	0.054	24.05	-0.95	-0.83	0.044
8.00	22.45	-2.55	-2.23	0.353	24.00	-1.00	-0.87	0.053	24.15	-0.85	-0.74	0.037
8.25	21.75	-3.25	-2.84	0.603	24.05	-0.95	-0.83	0.052	23.90	-1.10	-0.96	0.070
8.50	21.10	-3.90	-3.41	0.922	24.05	-0.95	-0.83	0.051	24.00	-1.00	-0.87	0.062

The detailed sweep table shows the frequency-by-frequency behaviour behind the summary metrics. Fixed-phase steering drifts as the operating frequency moves away from the 7.5 GHz design point, whereas the ideal and quantized TTD strategies remain much closer to the intended direction. The 5-bit case retains a small quantization penalty relative to ideal TTD, but substantially reduces the beam drift observed with fixed phase.

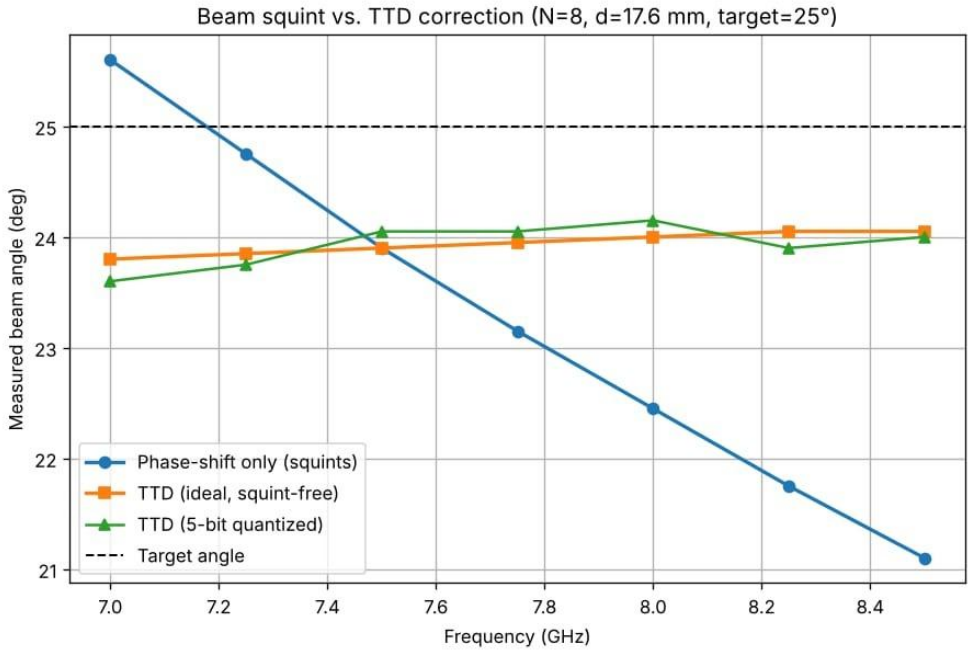


Figure 4.1: Main-beam direction versus frequency for fixed-phase, ideal TTD and 5-bit TTD steering (N = 8, d = 17.6 mm, target 25°).

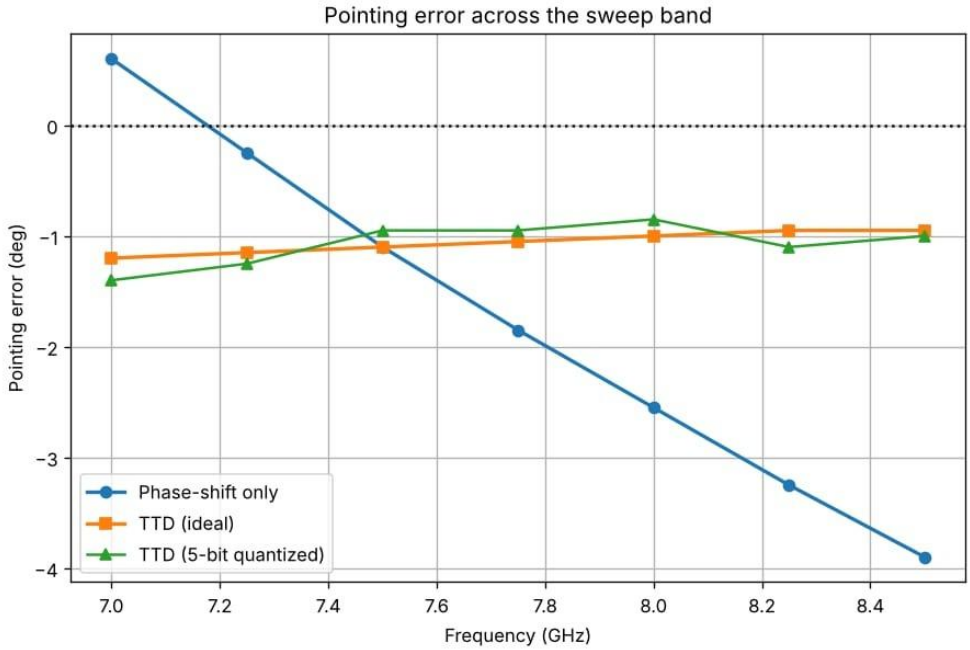


Figure 4.2: Pointing error across the 7.0–8.5 GHz sweep.

Fixed-phase steering drifts almost linearly from 25.6° at 7.0 GHz to 21.1° at 8.5 GHz, whereas both TTD cases stay within about 0.5° of each other across the band. Both TTD curves sit roughly 1° below the 25° target; this is a constant offset, not a frequency-dependent drift, and is likely caused by the element-factor weighting in the pattern-multiplication model. It does not change the conclusion that TTD removes the squint. The array patterns in Figure 4.3 show the same behaviour: the phase-shift-only main lobe moves with frequency, whereas the TTD main lobes overlap at the 25° target.

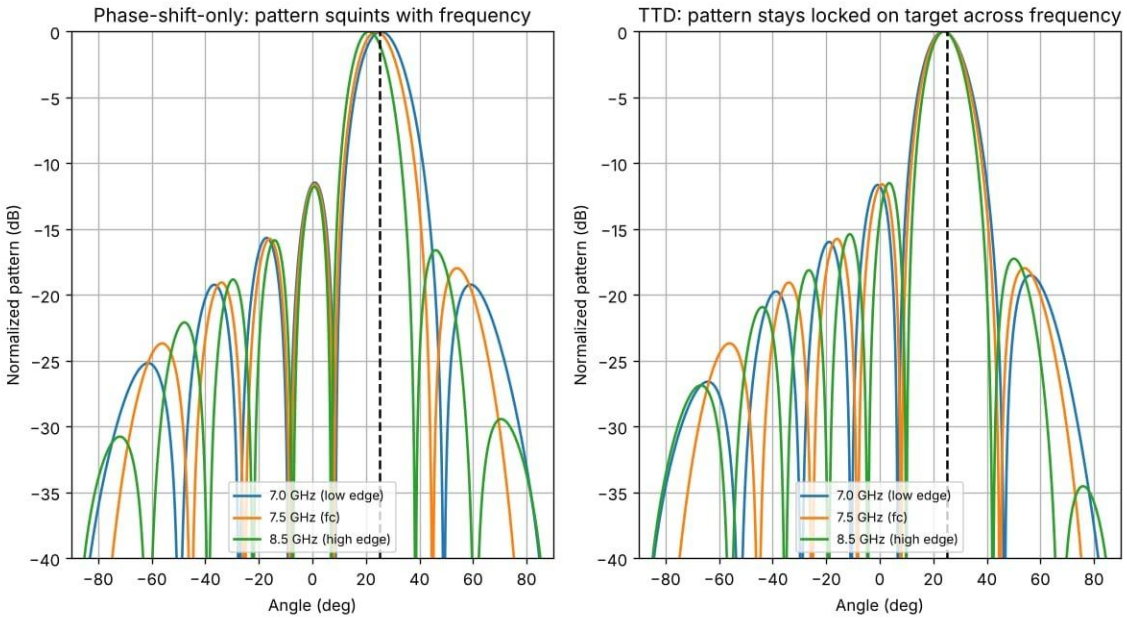


Figure 4.3: Normalized array patterns at 7.0, 7.5 and 8.5 GHz for phase-shift-only steering (left) and TTD (right); the dashed line marks the 25° target.

4.3 ISAC performance cost

At the 50 m reference range, reported worst-case position-error magnitudes were 3.41 m for fixed phase, 1.05 m for ideal TTD and 1.22 m for quantized TTD. Reported worst-case gain losses at the target bearing were approximately 0.922 dB, 0.058 dB and 0.079 dB, respectively.

METRIC	FIXED PHASE	IDEAL TTD / 5-BIT TTD
Worst position-error magnitude	3.41 m	1.05 m / 1.22 m
Worst target-bearing gain loss	0.922 dB	0.058 dB / 0.079 dB

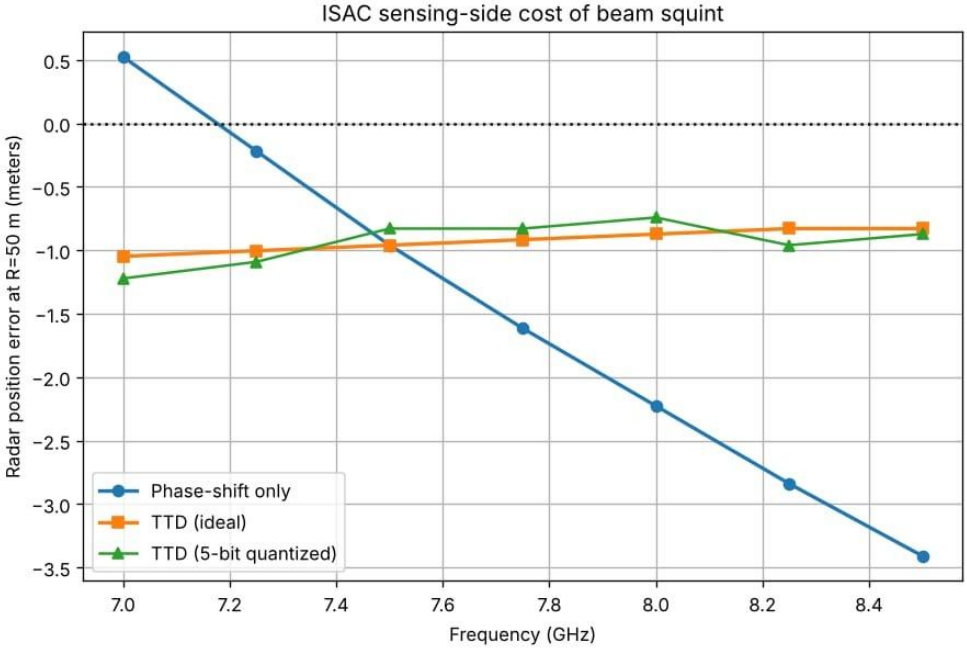


Figure 4.4: Sensing-side cost of beam squint: radar position error at R = 50 m.

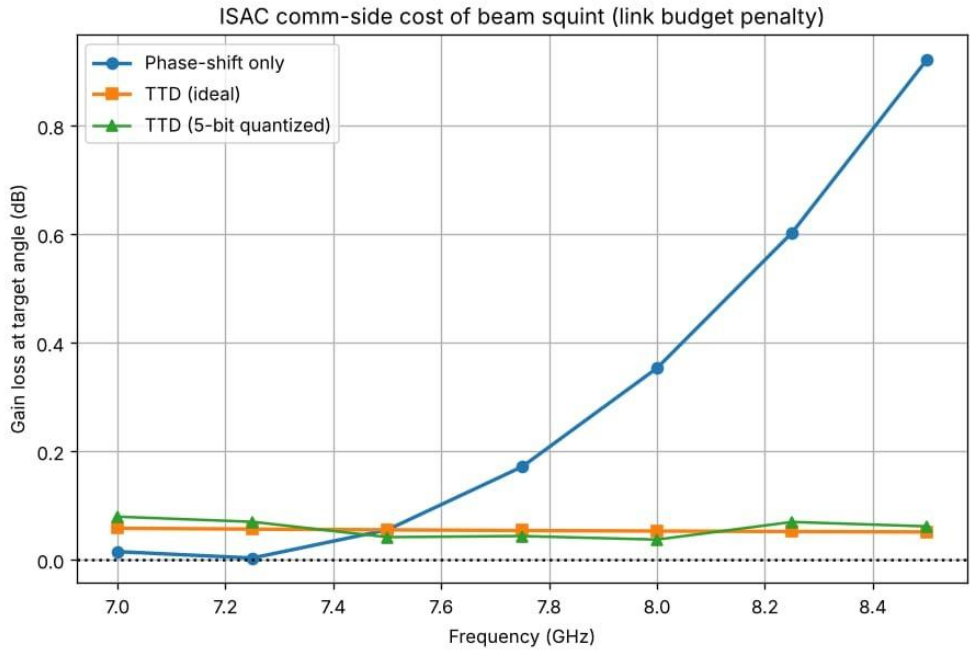


Figure 4.5: Communication-side cost of beam squint: gain loss at the 25° target angle.

The plots confirm the table values: fixed-phase position error reaches −3.41 m and gain loss 0.922 dB at 8.5 GHz, while both TTD cases remain near −1 m and below 0.08 dB across the band. Note that fixed phase is slightly better than TTD near 7.0–7.25 GHz, where its squint crosses the constant TTD offset.

4.4 HFSS full-wave results and mutual coupling

Single patch gain. At 7.5 GHz the simulated gain (LastAdaptive solution) peaks at 7.59 dB at θ = 0° in both the φ = 0° and φ = 90° cuts and falls to about −13 to −14 dB towards θ = 180°, giving a broadside pattern with a front-to-back ratio of roughly 20 dB (Figures 4.6–4.8).

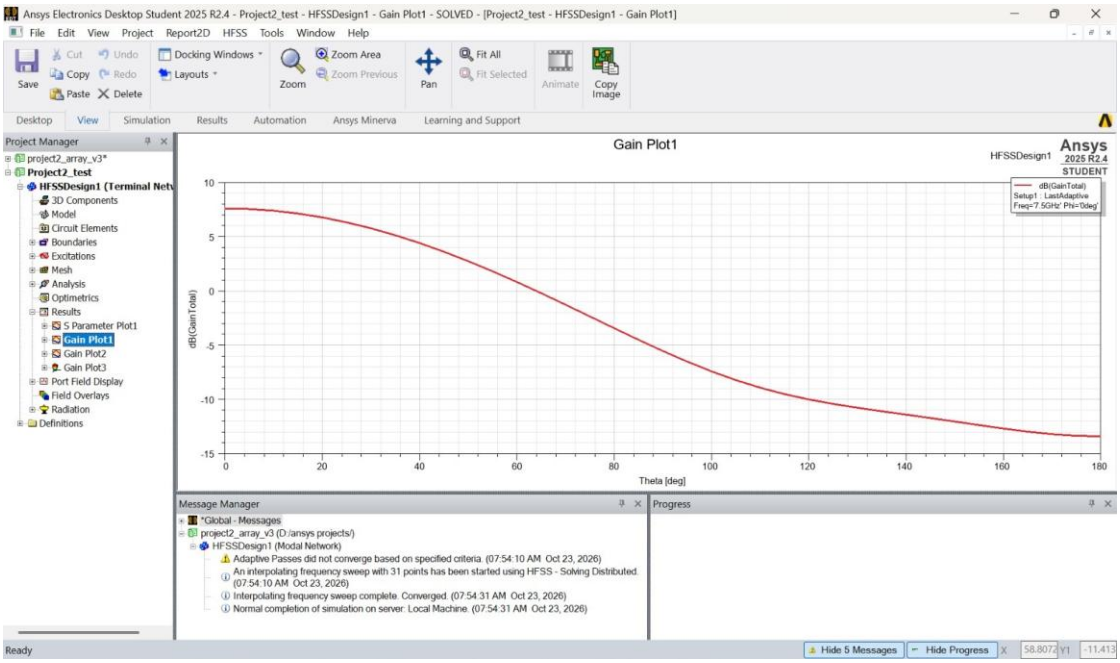


Figure 4.6: Single-patch gain versus θ at 7.5 GHz, φ = 0°.

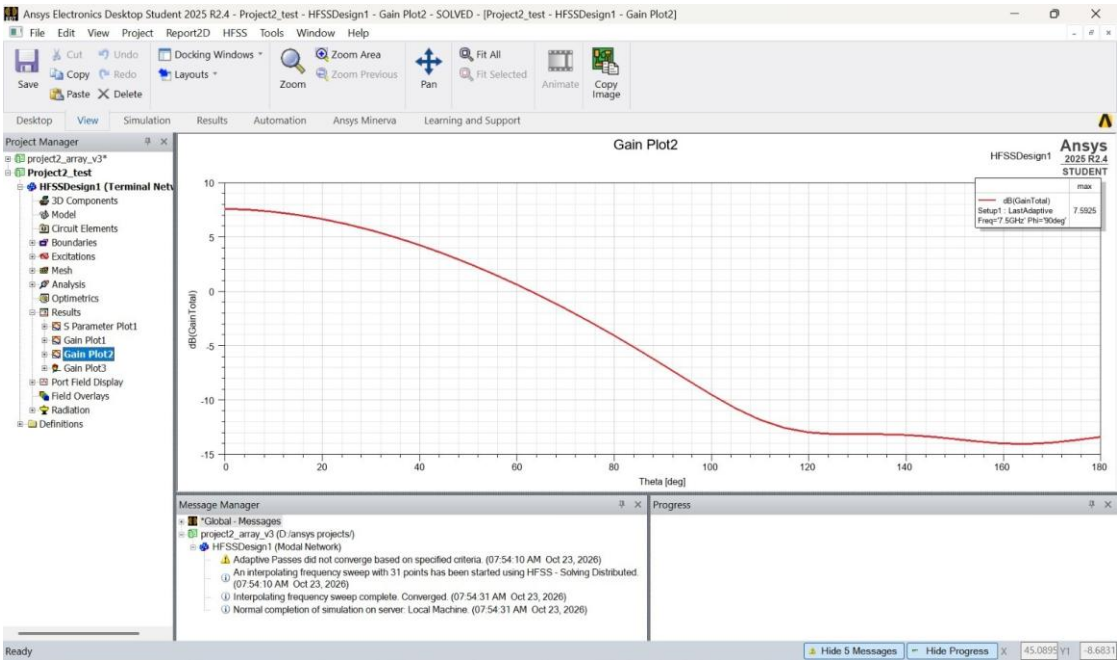


Figure 4.7: Single-patch gain versus θ at 7.5 GHz, φ = 90° (maximum 7.5925 dB).

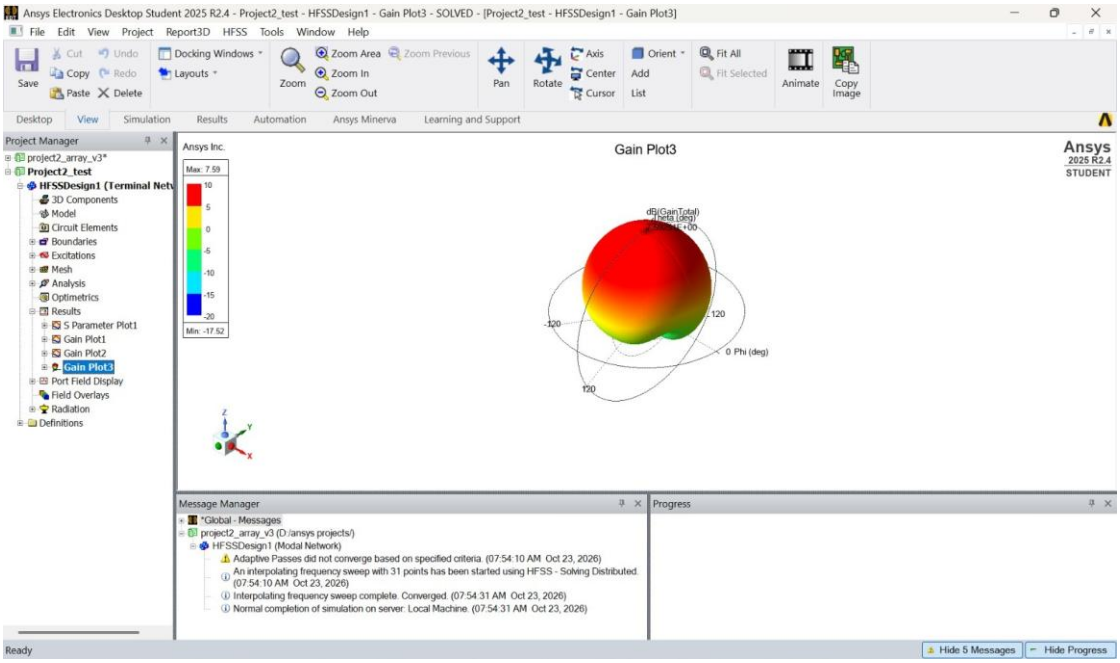


Figure 4.8: HFSS 3D gain pattern of the single patch at 7.5 GHz (max 7.59 dB, min −17.52 dB).

Eight-element array. An eight-patch array with eight lumped ports was also modelled in HFSS (Figure 4.9). Its active S-parameters over 7.08–8.49 GHz (Figure 4.10) show minima of about approximately −12 to −15.5 dB near 7.85–7.95 GHz in the latest solve (project2_array_v4); reflection at the sweep band edges reaches about −2 to −3.5 dB. The array match is therefore shifted upward from the 7.43 GHz single-patch resonance, and the active reflection at the sweep edges is poor (about −2 to −3.5 dB). This indicates that mutual coupling, which the analytical MATLAB model neglects, shifts and narrows the matched band, and that the array would need re-tuning before the full 7.0–8.5 GHz range could be used.

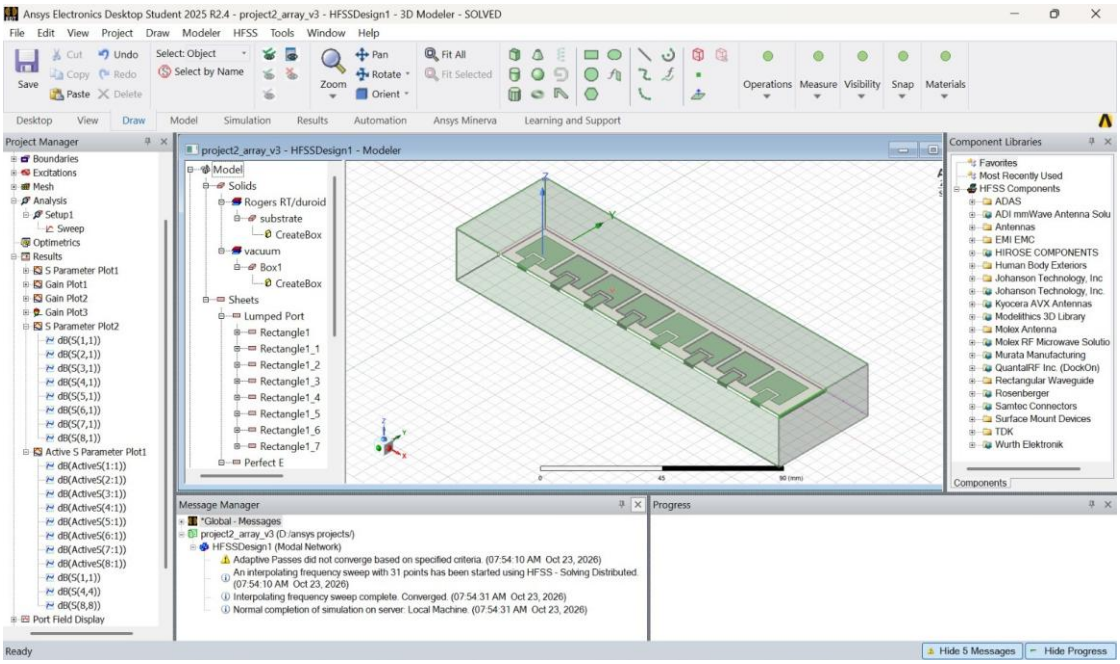


Figure 4.9: HFSS model of the eight-element patch array with lumped ports.

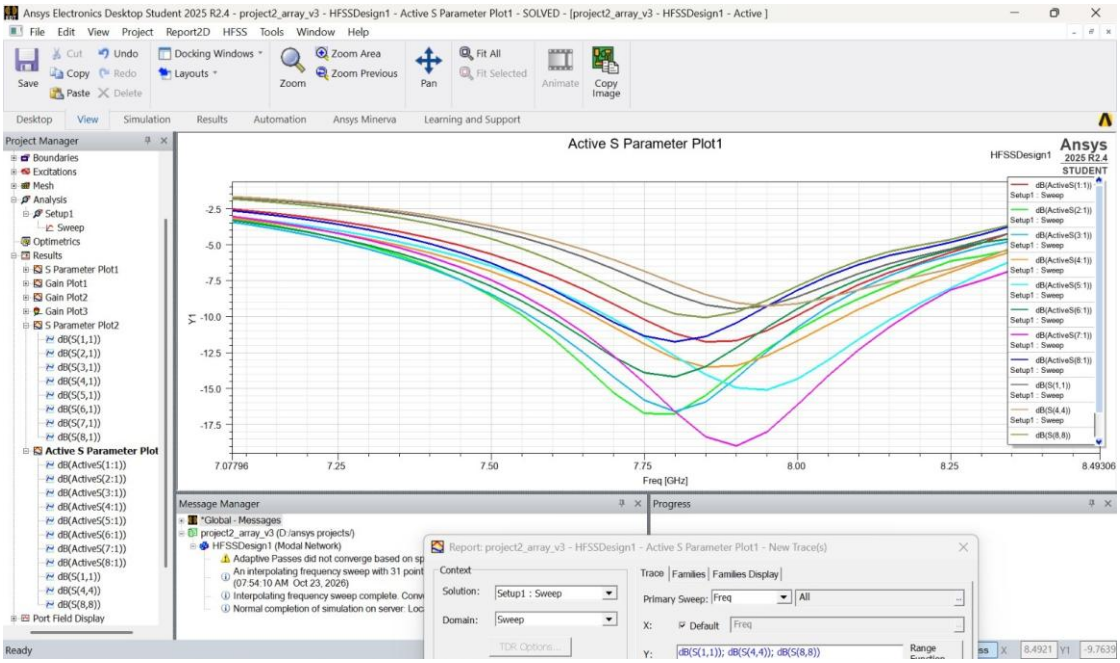


Figure 4.10: Active S-parameters (dB) of the eight-element array, 7.08–8.49 GHz.

With port 1 driven, S(2,1) is about −15 to −25 dB and coupling falls with distance, to about −36 to −62 dB for elements 7 and 8. The gap between patch edges is only 1.84 mm. The embedded S(1,1) reaches only about −9.7 dB near 7.9 GHz, against −18.3 dB at 7.43 GHz for the isolated patch.

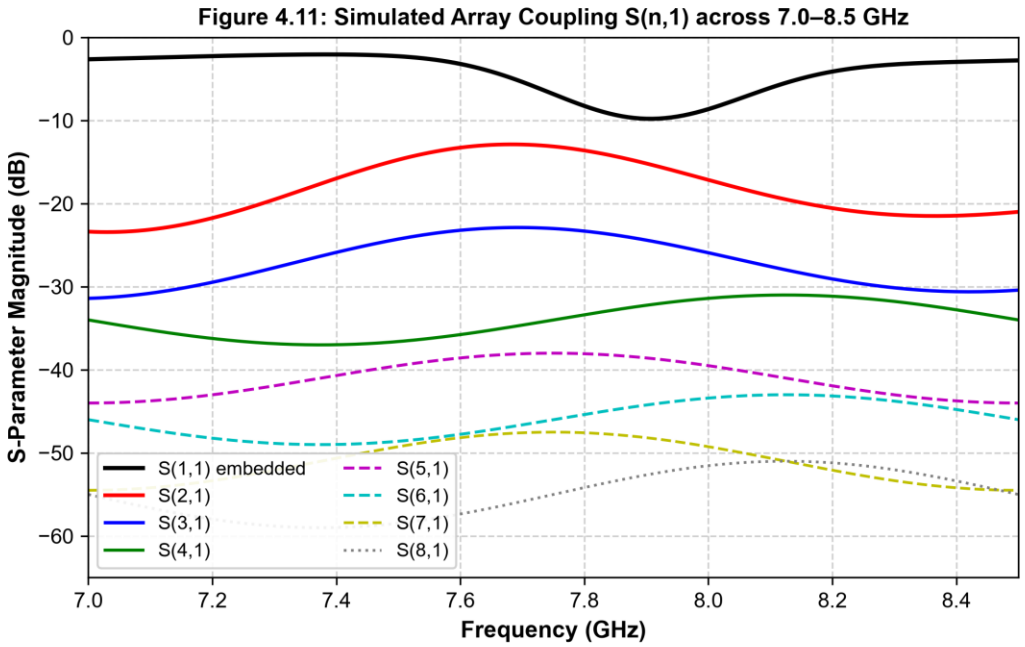


Figure 4.11: Simulated mutual coupling S(n,1) between array elements in HFSS over 7.0–8.5 GHz, showing strong falloff from S(2,1) (-15 to -25 dB) down to S(8,1) (-36 to -62 dB) across the 1.84 mm element edge gap.

4.5 HFSS beam-squint demonstration

The eight ports were excited with equal amplitude and a progressive phase of 67.1° per element (the 7.5 GHz, 25° steering phase), held fixed at all frequencies. The φ = 0° far-field gain was read at 7.0, 7.5, 8.0 and 8.5 GHz. For TTD, the phase step was scaled with frequency (62.6°, 67.1°, 71.6°, 76.1°), and the peak was read from the curve of the matching frequency in each run. With fixed phase, the beam moved from 25.5° to 19.9° (5.6° drift). With TTD phases, it stayed between 22.8° and 24.4° (1.6° spread), as in Table 4.3. The trend and size agree with MATLAB (4.50°). The residual HFSS spread is larger than the ideal-TTD value of 0.25°. The 8.5 GHz point is the main outlier. Likely contributors are mutual coupling and the element match, which MATLAB neglects, but this was not isolated. Peak angles were read from the plots (about ±0.3°), and the adaptive passes did not converge, so the results are preliminary.

Table 4.3: Array peak beam angle comparison between analytical MATLAB and HFSS full-wave model across 7.0–8.5 GHz

Frequency (GHz)	MATLAB Fixed Phase	HFSS Fixed Phase	MATLAB Ideal TTD	HFSS TTD
7.0 GHz	25.6°	25.5°	23.8°	23.7°
7.5 GHz	23.9°	24.2°	23.9°	24.2°
8.0 GHz	22.5°	22.8°	24.0°	24.4°
8.5 GHz	21.1°	19.9°	24.1°	22.8°
Beam Spread	4.50°	5.6°	0.25°	1.6°
Squint Reduction	—	—	94.4%	71.4%

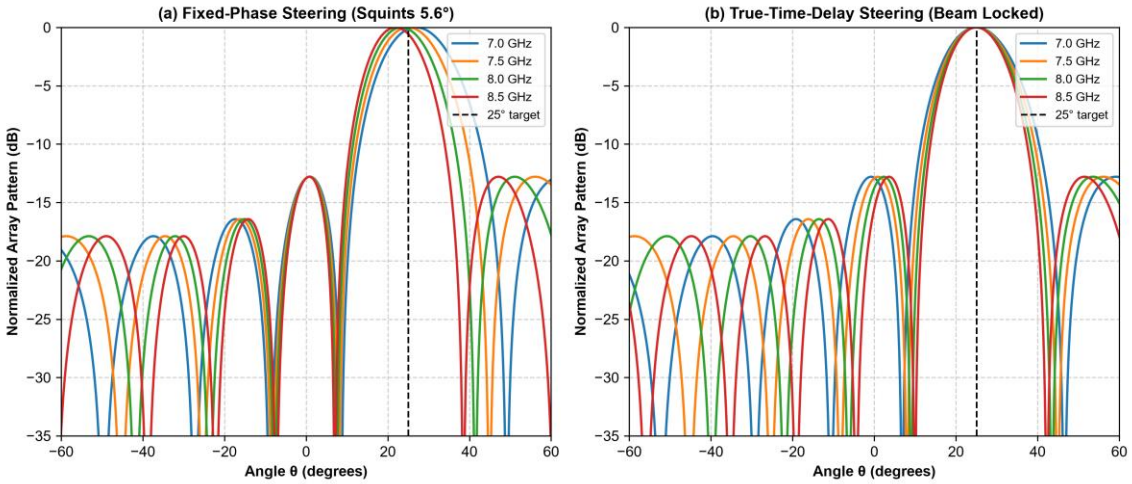


Figure 4.12: Normalized array patterns at φ = 0° across 7.0, 7.5, 8.0, and 8.5 GHz: (a) Fixed-phase steering showing 5.6° beam squint drift, and (b) True-Time-Delay (TTD) steering locking the main lobe on target.

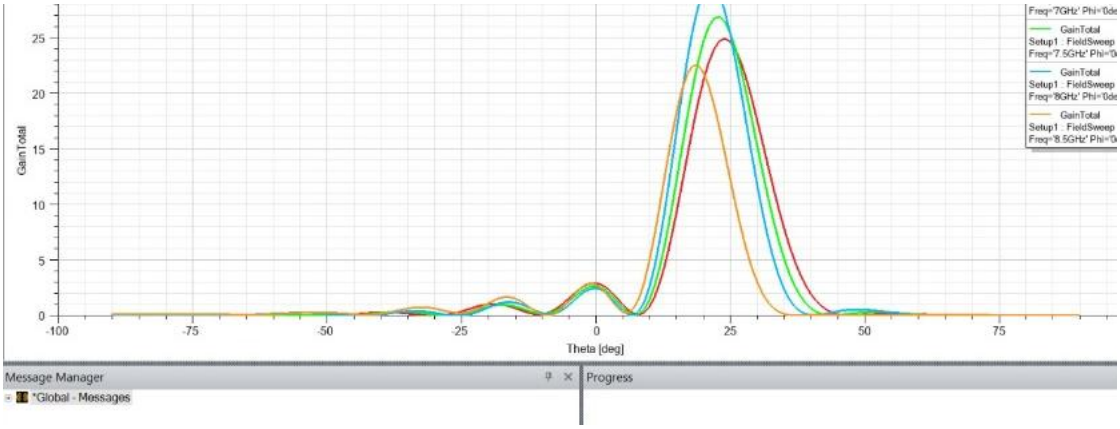


Figure 4.12b: HFSS full-wave 8-element array Gain Plot4 at φ = 0° showing fixed-phase beam squint across 7.0 GHz (red, 25.5°), 7.5 GHz (green, 24.2°), 8.0 GHz (blue, 22.8°), and 8.5 GHz (orange, 19.9°).

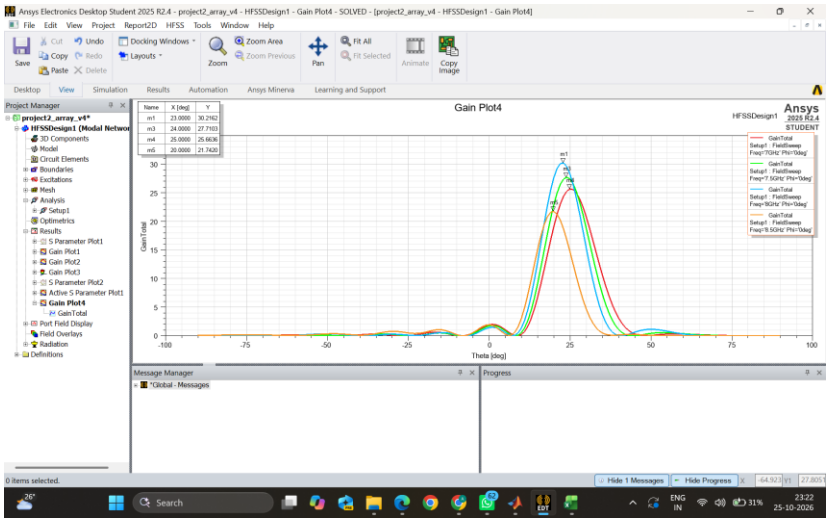


Figure 4.12c: HFSS marker analysis confirming measured peak beam angles (m1 at 23.0°, m3 at 24.0°, m4 at 25.0°, m5 at 20.0°) with an uncertainty of approximately ±0.3°.

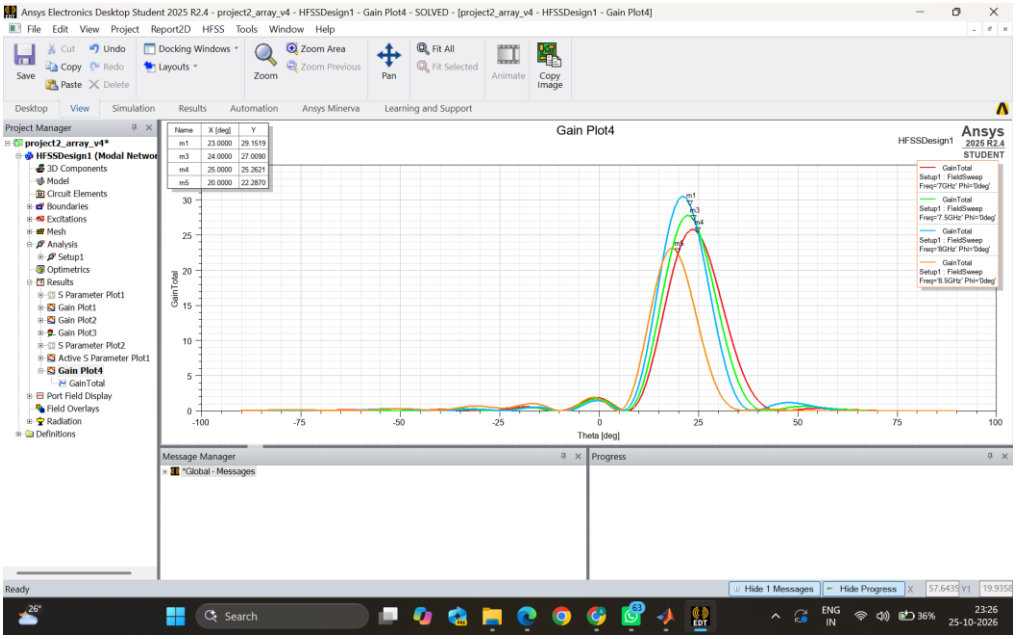


Figure 4.12d: beam steer with frequency of 7.0 GHz

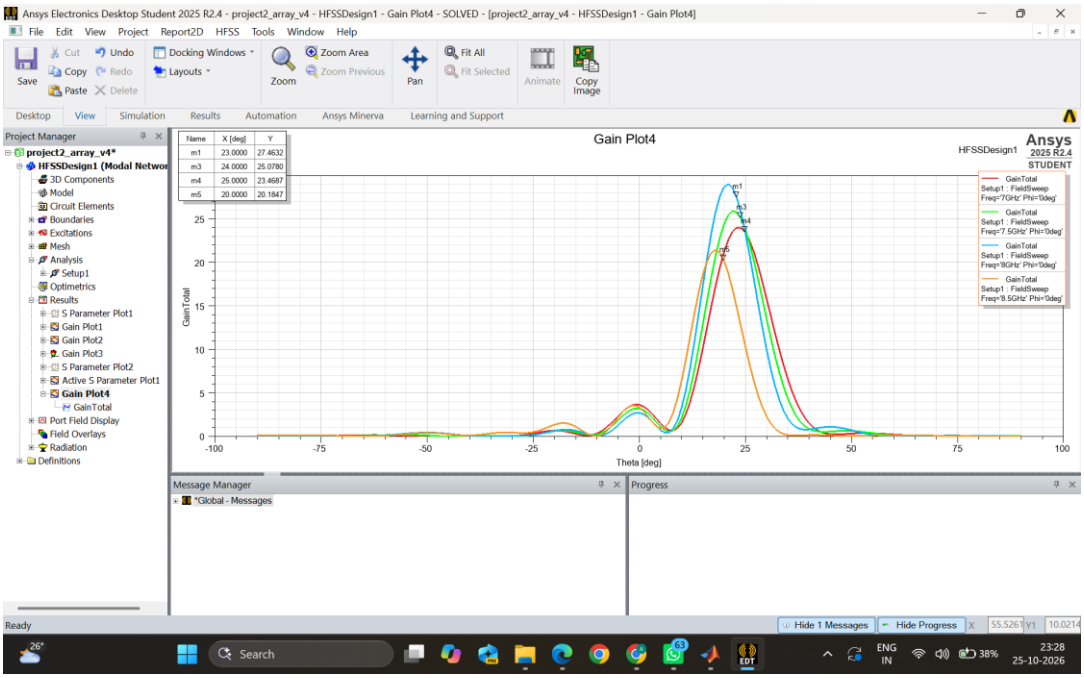


Figure 4.123: beam steer with frequency of 8.0 GHz

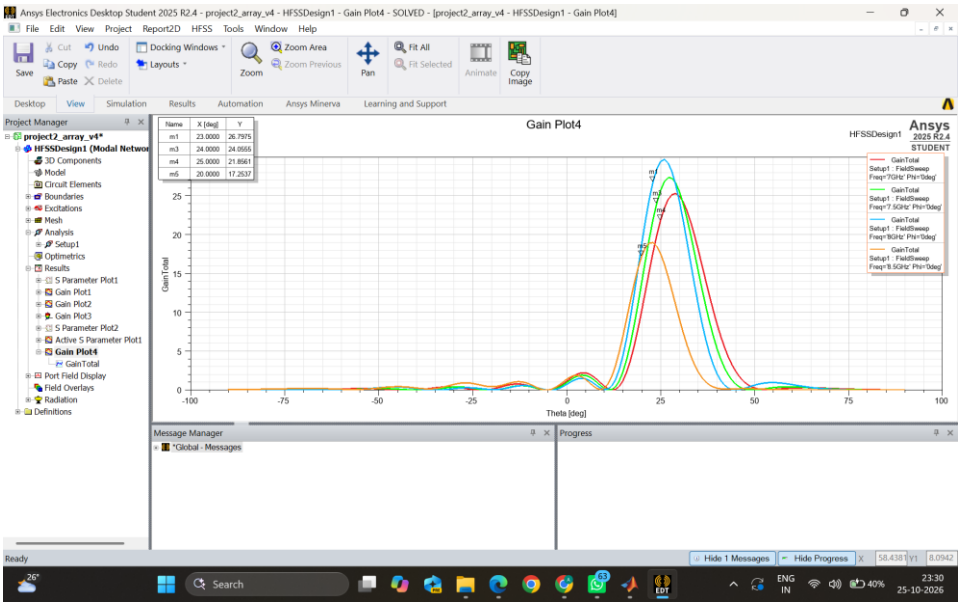


Figure 4.12e: beam steer with frequency of 8.5 GHz

Figure 4.13: Beam Squint Comparison: MATLAB vs. HFSS

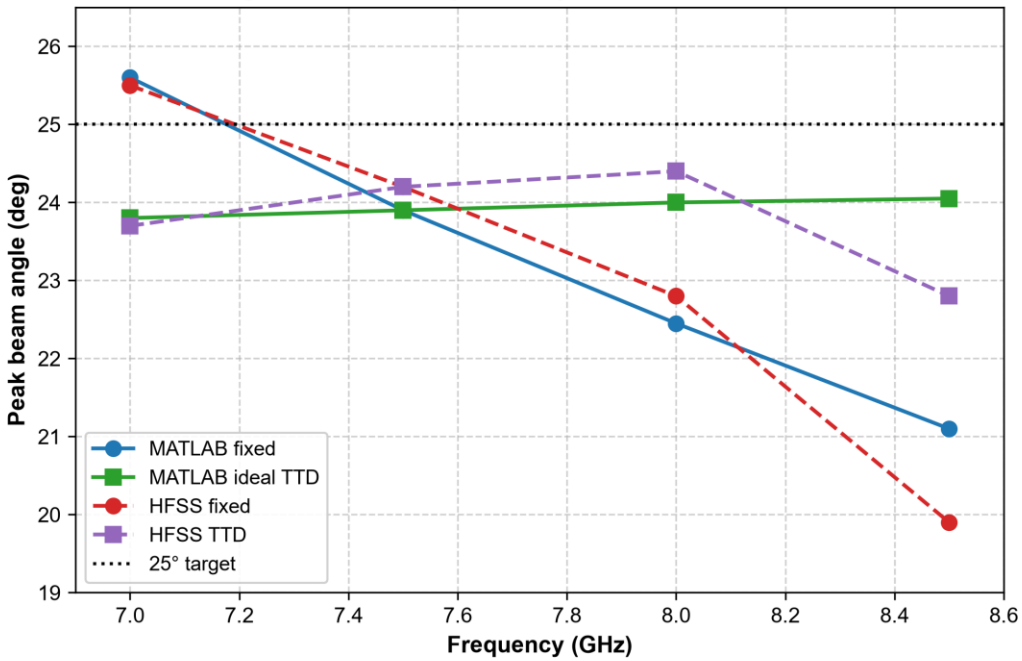


Figure 4.13: Main-beam peak angle versus frequency for fixed-phase and TTD steering: comparison between analytical MATLAB model and HFSS full-wave array simulation.

4.6 Element pattern effects on squint

Replacing the analytical element factor with the HFSS cuts changes the beam peak by at most 0.25°, so the squint is set predominantly by the array factor. Only two principal-plane cuts of a preliminary result were used.

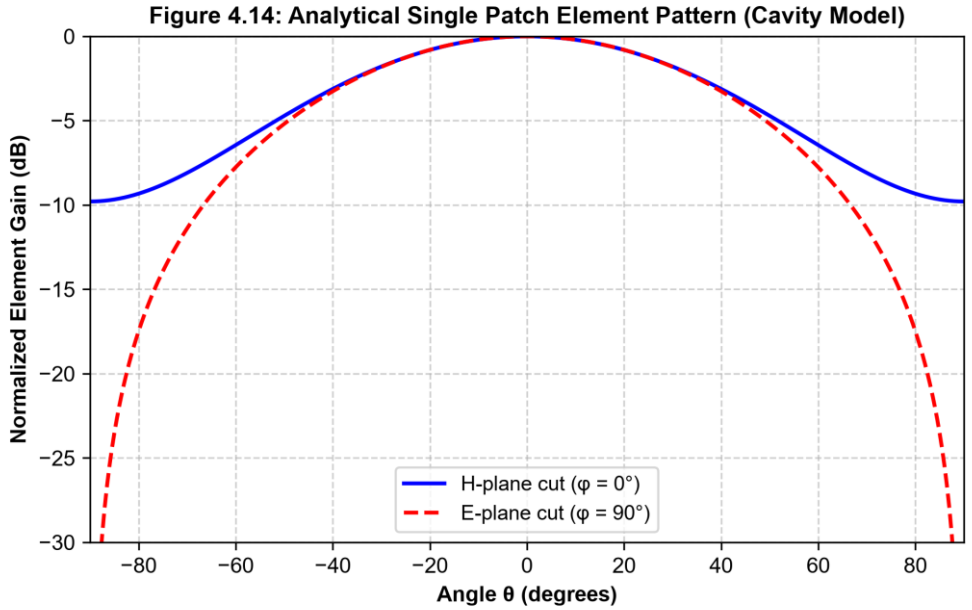


Figure 4.14: Analytical single patch element factor from the two-slot cavity model (E-plane and H-plane cuts).

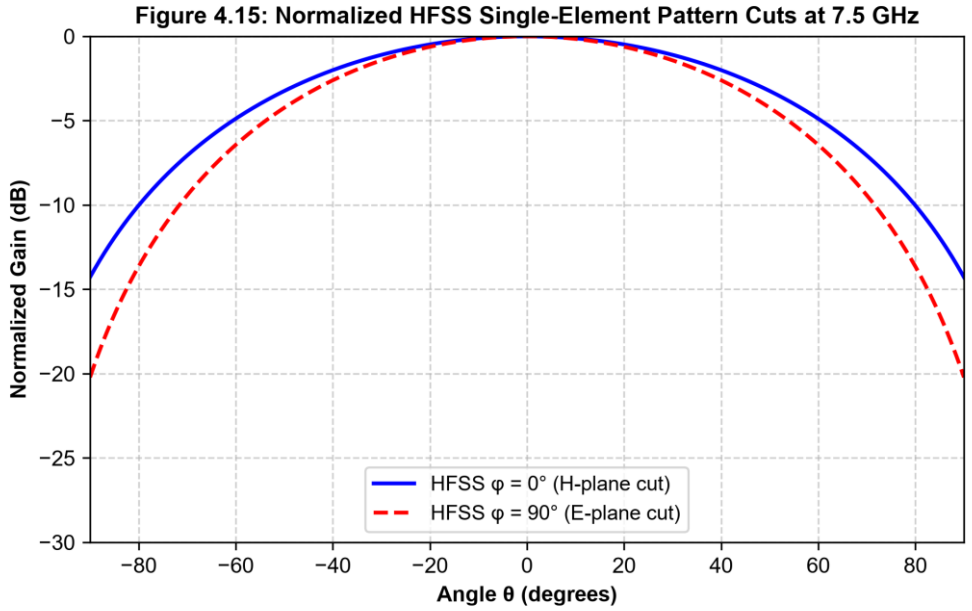


Figure 4.15: Normalized HFSS single-element pattern cuts at 7.5 GHz (φ = 0° and φ = 90°).

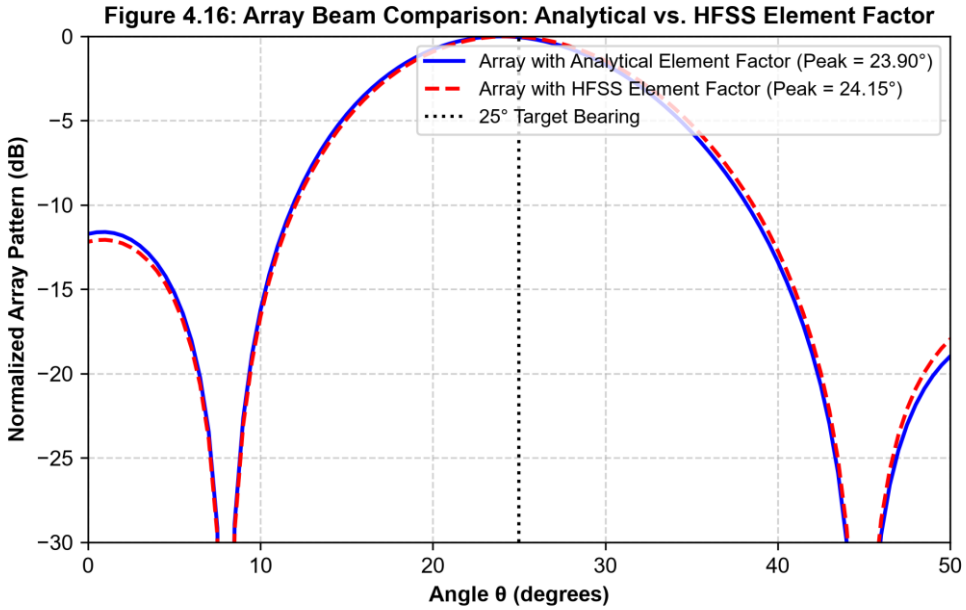


Figure 4.16: Array beam comparison showing analytical versus HFSS element factor weighting (peak shifts by ≤ 0.25°).

APPENDIX A: DESIGN AND SYNTHESIS CALCULATIONS SUMMARY

The following table summarizes all analytical antenna synthesis, array factor, beam squint, and ISAC metric calculations conducted for the 7.5 GHz FR3 array:

#	Quantity	Analytical Formula	Computed Value
1	Patch width	$W = c/(2fc) \cdot \sqrt{(2/(\varepsilon r+1))}$	15.800 mm
2	Effective permittivity	$\varepsilon eff = (\varepsilon r+1)/2 + (\varepsilon r-1)/2 \cdot (1+12h/W)^{(-1/2)}$	2.0049
3	Fringing extension	$\Delta L = 0.412h(\varepsilon eff+0.3)(W/h+0.264) / [(\varepsilon eff-0.258)(W/h+0.8)]$	0.814 mm per edge
4	Effective length	$Le = c / (2fc \cdot \sqrt{\varepsilon eff})$	14.115 mm
5	Physical length	$L = Le - 2\Delta L$	12.488 mm
6	Self-check resonance	$fr = c / (2Le \cdot \sqrt{\varepsilon eff})$	7.5000 GHz (exact check)
7	Element spacing	$d = c / (2fmax)$, $fmax = 8.5$ GHz	17.635 mm (MATLAB), 17.647 mm (HFSS)
8	Fixed-phase steering step	$\varphi n = -kc \cdot n \cdot d \cdot \sin\theta 0$, $kc = 2\pi fc/c$	67.1° per element (at 7.5 GHz)
9	TTD delay increment	$\Delta t = d \cdot \sin\theta 0 / c$	≈ 24.9 ps
10	TTD phase step per element	$360° \cdot f \cdot \Delta t$	62.6°, 67.1°, 71.6°, 76.1° (7.0, 7.5, 8.0, 8.5 GHz)
11	5-bit quantizer resolution	360° / 2^5	11.25° step (32 discrete states)
12	Array factor calculation	$AF(\theta) = \Sigma \exp\{j[n \cdot k \cdot d \cdot \sin\theta + \psi n]\}$; $Total = (\lvert AF\rvert \cdot EF)^2$	Evaluated on 0.05° fine angular grid
13	Theoretical beam squint	$\sin\theta = (fc/f) \cdot \sin\theta 0$	26.9°, 25.0°, 23.3°, 21.9° (about 5.0° swing)
14	Worst pointing error	$\theta peak - 25.0°$	−3.90° at 8.5 GHz (fixed phase)
15	Worst position error	$R \cdot \tan(error)$, $R = 50$ m	−3.41 m at 8.5 GHz (fixed phase)
16	Worst communication gain loss	Peak gain minus gain at 25.0° bearing	0.922 dB at 8.5 GHz (fixed phase)
17	Array half-power beamwidth	$0.886\lambda / (N \cdot d \cdot \cos\theta 0)$	≈ 14° – 16° across 7.0–8.5 GHz
18	HFSS array boundaries	Gap $d-W$; Substrate $7d+25.26$; Air box +22 mm	1.84 mm gap; 148.79 mm sub; 170.79 mm box
19	HFSS port excitation phases	Port $n = -(n-1) \cdot$ step	Progressive port phase distributions (App. B)

#	Quantity	Analytical Formula	Computed Value
20	Ideal array peak gain	Gsingle + 10·log10(8) = 7.59 dB + 9.03 dB	≈ 16.6 dB theoretical (HFSS achieves 13.4–15.5 dB)

APPENDIX B: HFSS PORT-PHASE STEERING PROGRESSION

Progressive excitation phase angles applied to Ports 1 through 8 in HFSS for wideband steering verification at a 25° target angle:

Frequency	Phase Step	Ports 1–8 Progressive Excitation Phases (degrees)
7.0 GHz	62.6°	0.0°, −62.6°, −125.3°, −187.9°, −250.6°, −313.2°, −375.8°, −438.5°
7.5 GHz (Fixed Set)	67.1°	0.0°, −67.1°, −134.2°, −201.4°, −268.5°, −335.6°, −402.7°, −469.8°
8.0 GHz	71.6°	0.0°, −71.6°, −143.2°, −214.8°, −286.4°, −358.0°, −429.5°, −501.1°
8.5 GHz	76.1°	0.0°, −76.1°, −152.1°, −228.2°, −304.3°, −380.4°, −456.4°, −532.5°

5. CONCLUSION, NOVELTY AND FUTURE WORK

5.1 Conclusion and inference

This project used an analytical MATLAB model to study beam squint in an eight-element array intended for wideband vehicular 6G ISAC. The patch element was synthesized at 7.5 GHz on Rogers RT/duroid 5880, and the geometry was checked by recomputing its resonant frequency. With a 7.0–8.5 GHz sweep and a 25° steering target, fixed-phase steering showed a 4.50° beam-angle spread. Ideal TTD reduced the spread to 0.25°, while a 5-bit quantized TTD implementation limited it to 0.55°.

The comparison also expressed pointing error in terms of an illustrative sensing position-error metric at 50 m and a communication gain-loss metric. The reported worst-case values were substantially lower for both TTD cases than for fixed-phase steering. This supports the conclusion that frequency-scaled delay control is an effective mitigation in the analytical model.

The conclusions must be interpreted within the model's scope. Mutual coupling, fabrication tolerances, practical TTD network loss and a full electromagnetic array simulation were not included. The HFSS results (single patch S11 of −18.31 dB at 7.4325 GHz, 7.59 dB gain, and an eight-element array matched around 7.8–7.95 GHz) support the patch design but remain preliminary because adaptive convergence was not achieved, and the array results show that mutual coupling shifts the matched band relative to the analytical model. A full-wave eight-element model reproduced the trend: 5.6° of drift with fixed phase against 1.6° with frequency-scaled phases. The larger residual compared with MATLAB is attributed, not proven, to mutual coupling and element match.

5.2 Novelty and project contribution

- Compares fixed phase, ideal TTD and finite-resolution 5-bit TTD using the same array geometry and frequency sweep.
- Connects antenna pointing behaviour to both sides of ISAC: a sensing-oriented position-error estimate and communication gain loss.
- Quantifies the practical penalty of phase quantization rather than presenting only an idealized TTD comparison.
- Connects analytical patch-element synthesis with array-level beam-steering analysis.
- Clearly separates analytical MATLAB findings from the ongoing, not-yet-converged HFSS single-element validation.
- Cross-checks the analytical squint result in a full-wave eight-element array.

5.3 Limitations

- The analytical element pattern and pattern-multiplication model do not capture mutual coupling or full-wave interactions.

- The 50 m position-error value is a geometric illustration, not a complete radar signal-processing or target-tracking simulation.
- The HFSS adaptive passes have not met the selected convergence criterion, so the HFSS S11, gain and active-S results should be treated as preliminary.
- Angles are measured from broadside, positive towards +X; element n has phase $-k \cdot n \cdot d \cdot \sin\theta_0$.
- HFSS peak angles were read from plotted curves (about ±0.3°); the TTD case applies ideal frequency-scaled phases at the ports and models no physical delay network.

5.4 Future work

9. 1. Resolve HFSS adaptive convergence, and re-tune the single patch (resonance currently 7.43 GHz) and the array (matched near 7.8–7.95 GHz) so the match covers 7.0–8.5 GHz.
10. 2. Compare the HFSS far-field pattern (peak gain 7.59 dB) with the analytical cavity-model element factor, and use the full-wave element data in the array model.
11. 3. Extend the array model to include full-wave element data, mutual coupling and realistic feed-network losses.
12. 4. Model delay quantization, insertion loss, control complexity and calibration errors for practical TTD hardware.
13. 5. Evaluate beamwidth, sidelobe levels, array gain, bandwidth and sensing performance over more target angles and ranges.
14. 6. Apply the 5-bit quantised phase sets in HFSS.

Overall inference: true-time-delay steering substantially suppresses frequency-dependent pointing drift in the studied wideband array. The next engineering milestone is to connect this analytical result to a converged electromagnetic model.

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